IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 3:

Match the filter with its purpose:

Mean Filter

Median Filter

Laplacian

Gaussian Filter

Purpose Options:

A. Removes salt & pepper noise

B. Image sharpening

C. Smooths image

D. Noise reduction with weighting

Write correct matches:

Mean → C. Smooths image

Median → A. Removes salt & pepper noise

Laplacian → B. Image sharpening

Gaussian → D. Noise reduction with weighting

Which filter preserves edges better — Mean or Median? Why?

Median filter preserves edges better than the mean filter. The median filter replaces each pixel with the median value of its neighboring pixels, which effectively removes salt-and-pepper noise without significantly blurring sharp edges. In contrast, the mean filter calculates the average of neighboring pixels, which can smooth the image but also blur edges and fine details. Therefore, the median filter is generally preferred when noise removal and edge preservation are both important.

One of the major advantages of the median filter is its ability to **preserve important edges while removing unwanted noise**. Edges represent boundaries between different objects or regions in an image and are important for later processes such as segmentation and feature extraction. Since the median filter does not calculate an average, it does not create as much intensity mixing across an edge as the mean filter does. Consequently, sharp boundaries are maintained more effectively.

For example, consider an image containing a black object against a white background. At the boundary between the object and the background, there is a large difference in pixel intensity. A mean filter averages the pixels on both sides of this boundary, producing intermediate gray values. This gradually changes the intensity and makes the boundary appear blurred. A median filter is less affected by these extreme differences and can remove isolated noisy pixels while maintaining a relatively sharp transition between the two regions.

Therefore, **the Median Filter is generally better than the Mean Filter for edge preservation**. The mean filter is suitable when the main objective is to obtain a smooth image, but it may sacrifice edges and fine details. The median filter provides a better balance between **noise removal and edge preservation**, making it especially useful for images affected by salt-and-pepper noise.

In conclusion, the Median Filter preserves edges better because it uses the median value instead of averaging neighboring pixels. It removes isolated noise effectively while maintaining sharp boundaries and important image details.